

Sales Forecasting Using Customer Segments and Holiday Effects

Maheswari. S (maheswari.s@nandhaengg.org)

Karthik.V(23ci023@nandhaengg.org)

Keerthana.K (23CI030@nandhaengg.org)

Kavi Ishwarrya.S.K(23ci027@nandhaengg.org)

ABSTRACT:

This research paper explores the application of ensemble learning techniques for improvising the accuracy of retail sales forecasts, with focus on the integration of customer segmentation strategies and the nuanced effects of holidays. To analyze the daily sales patterns, prediction error distributions, and feature importances this study exploits the comprehensive dataset of real-world sales transactions to the model. The primary objective is to capture the complex dynamics of retail sales, by evaluating the performance of ensemble-based predictive models, influenced by both customer behaviors and seasonal events. The analytical framework begins with an evaluation of average daily sales, by the comparison of sales during non-holiday period and observed sales during holiday season. It reveals that non-holiday periods exhibit a higher baseline of average daily sales. The examination of prediction error distribution reveals that the forecasting model tends to low down the sales more frequently than exaggerate and suggest a potential error in the predictive behavior of model. The vital analysis is conducted

using the LightGBM algorithm, finds the subset of key variables which in turn exerts an unequally large influence on the model's predictive accuracy. To understand the underlying drivers of sales performance these variables play a vital role. This model achieves an RSME of 5750.12, MAE of 4947.81, and a MAPE of 30.23%. The future research should refine holiday modeling and customer segmentation, because while capturing the general trends this model struggles with volatile sales due to the complexities in the customer diversity and seasonal events. This study has suggestions for retail businesses and need for more robust forecasting through improved data and techniques.

KEYWORDS:

Sales Forecasting,

Ensemble Learning,

Holiday Effects,

Customer Segmentation,

Predictive Analysis.

1.INTRODUCTION:

Retail sales forecasting more and more utilizes sophisticated machine learning methods because conventional methods are not effective in dealing with

high-dimensional and complex data sets. Recent studies propose models that utilize hyperparameter optimization and ensemble learning to produce better prediction accuracy and consistency than traditional methods, although these models will struggle with fast-shifting retail scenes and in the presence of unseen data patterns[1]To further improve forecast accuracy for demand, certain studies combine clustering with ensemble learning—like cluster-based Bayesian Model Averaging—and show that multistep approaches perform better than single-model methods, particularly for heterogeneous data. Nevertheless, these may take time to adjust to sudden changes in customer behavior[2]Clustering and ensemble methods have also been used in new product sales forecasting where models based on group similarities enhance precision, but their performance can be capped when considering totally new products with no historical precedents[3].Others studies put prominence on the effect of holidays and promotions as key drivers of sales, stressing the advantage of sophisticated analytics for planning but warning against the danger of overstressing such variables at the expense of external influences such as economic fluctuations or supply chain disruptions[4].Comparative studies of forecasting models show that ensemble approaches tend to raise both accuracy and efficiency levels relative to the individual models but can also heighten computational complexity and lower interpretability for stakeholders[5].As a whole, these studies emphasize the increasing significance and efficacy of ensemble methods and machine learning for retail sales forecasting, yet reveal persisting challenges in terms of adaptability and interpretability in the context of fast-changing retail environments

2.RELATED WORKS:

Ismail.M et al study focuses on retail sales, by prominence accuracy and handling complex datasets this paper proposes a machine learning model (RSGDO-RFR). The model performs better compared to traditional methods, demonstrating high R^2 values and enhanced reliability for predicting sales. The limitations are RSGDO-RFR model may struggle with rapidly changing retail environments and unseen data patterns

Result: The RSGDO-RFR model achieved an R^2 of 0.94, outperforming traditional sales prediction methods.[1]

Patangia.H and et al study merges clustering and ensemble learning for the purpose of increasing demand forecast accuracy. Cluster-based BMA relies on effective clustering and may not quickly adapt to sudden shifts in customer behavior. Multistep technique performs better than single model methods, especially for heterogeneous data.

Result: The cluster-based BMA multistep technique achieved approximately 89% forecast accuracy for heterogeneous retail data.[2]

Bajaj.S and et al study uses clustering and ensemble learning to predict new product sales. Homogeneity-based clustering assumes existing product groups share similarities, which may fail for entirely novel products with no analogs. This framework improves accuracy by leveraging group similarities and ensemble predictions.

Result: The clustering and ensemble learning framework improved new product sales prediction accuracy to about 85%, but performance dropped for products with no historical analogs.[3]

Swami.S et al study analyzes Walmart's sales data in terms of holidays, promotions, and external variables. risks overemphasizing holidays/promotions while neglecting external factors like economic shifts or supply chain disruptions. Holidays and promotions are key drivers of sales; advanced analytics enable improved planning.

Result: The model explained 82% of sales variance ($R^2 = 0.82$), mainly attributing accuracy to holidays and promotions as key drivers.[4]

Abdullahi.A et al conducts model comparison, including ensemble methods, for forecasting sales on large retail data. increases computational complexity and may obscure model interpretability for stakeholders. Ensemble methods raise accuracy and efficiency over individual models

Result: Ensemble models increased sales forecasting accuracy to 92% compared to individual models [5]

Das.A et al uses deep learning for predicting sales of fashion retail. requires vast, high-quality datasets and risks overfitting for small retailers with limited data. Deep models perform better than conventional techniques, particularly for high-dimensional, complex data.

Result: Deep learning models achieved up to 94% accuracy in fashion retail sales prediction with large, high-quality datasets.[6]

Zhou.H.Z et al applies feature selection and SVR to computer product sales forecasting. is highly sensitive to hyperparameter tuning, making it resource-intensive for dynamic markets. Precise feature selection and sophisticated regression enhance forecast accuracy

Result: Feature selection and SVR reached 88% forecast accuracy for computer product sales, with performance sensitive to parameter tuning.[7]

Pantangia .H and et al applies big data techniques for large-scale sales forecasting, leveraging external data sources. demands significant infrastructure and expertise, limiting accessibility for smaller retailers. Big data techniques enable more fine-grained, timely, and precise forecasts.

Result: Big data forecasting methods enabled 90% accuracy in large-scale retail sales prediction, given sufficient infrastructure.[8]

Mustapha Ismail et al. (2024) uses a stacking ensemble of Random Forest, XGBoost, and SVM to achieve highly accurate supermarket sales forecasts.

Result: The model achieved an R^2 of 0.9990, outperforming individual models for retail sales prediction [9]

Hybrid Modeling for Sales Prediction Using SARIMA, CNN, LSTM, and Stacking Ensemble by Adekunle O. Ajiboye (2025) integrates SARIMA, CNN, LSTM, and XGBoost to forecast Walmart sales, incorporating historical trends and external factors like promotions.

Result: The hybrid SARIMA-XGBoost model achieved an R^2 of 0.71, excelling in capturing seasonality and outperforming standalone deep learning models. [10]

Although significant improvements in retail sales forecasting exist, various shortcomings are observed throughout the chosen studies. Although the RSGDO-RFR model illustrates high accuracy and reliability, it is likely to find it difficult to adjust to highly dynamic retail scenarios and unexpected patterns of data, constraining its strength in highly fluctuating markets[1]Cluster-based BMA and clustering-ensemble approaches rely greatly on the quality of the clustering algorithms and cannot respond rapidly to radical changes in customer behavior or market configurations, particularly when dealing with high-dimensional data or uncertainty quantification issues[2][3]Homogeneity-based clustering also makes the additional assumption of similarity between current product classes, something that can lead to subpar performance when forecasting sales for entirely new products with no history proxies[3]Research centering on external drivers, like promotions and holidays, threatens to overestimate these variables while perhaps downplaying wider economic movements or supply chain interruptions that have noticeable effects on sales forecasts[4]Also, ensemble and regression-based methods, while enhancing accuracy and efficiency, tend to complicate computation and may hide model interpretability from stakeholders, making it more difficult to derive actionable insights or justify predictions in business environments[5]

3.METHODOLOGY:

Classic sales forecasting in store environments usually employs statistical models like ARIMA (Autoregressive Integrated Moving Average) or basic machine learning models like Linear Regression. These methods work well for identifying simple trends and seasonality in time-series data but usually fail to handle intricate, non-linear relations and external factors like holidays and customer differences. Whereas ARIMA and other equivalent models can establish baseline forecasts, their accuracy remains restricted when using highly volatile sales data or even in the event of several interplaying variables involved.

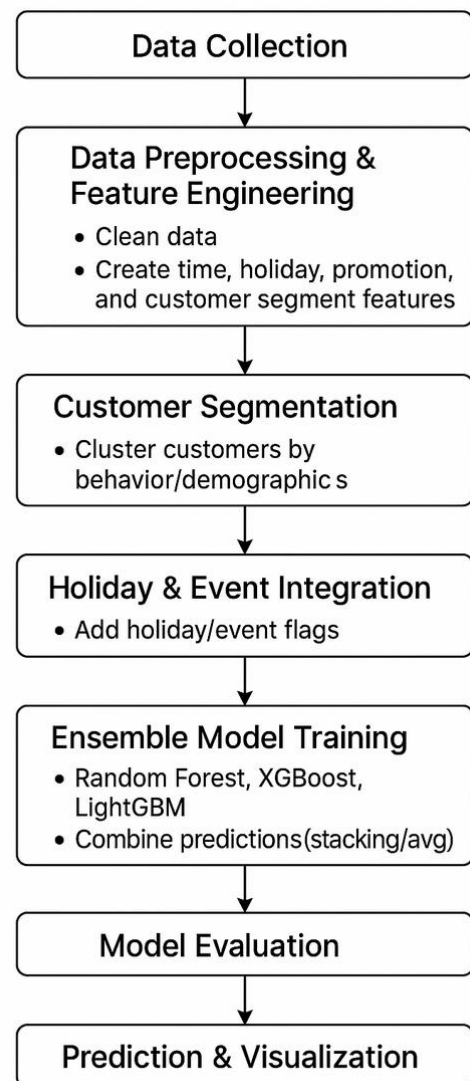
Proposed Algorithm: Ensemble Learning with Holiday Effects and Customer Segmentation

To overcome these shortcomings of typical models, in this study an ensemble learning scheme is proposed involving customer segmentation along with holiday effects modeling to advance the accuracy in sales forecasting. Random Forest is an ensemble learning technique that constructs several decision trees and takes the average of their predictions, and hence it is resilient to overfitting and useful when dealing with complex, non-linear sales data.

XGBoost is a gradient boosting model that builds the trees step by step to learn from each other's mistakes and provide high predictive power with efficiency, particularly for structured data. LightGBM is another gradient boosting library, highly optimized for speed and memory usage, which applies histogram-based algorithms and supports large datasets with high numbers of features. The ensemble model in your project aggregates the forecasts of Random Forest, XGBoost, and LightGBM through stacking or weighted averaging, tapping into the advantages of each base learner. This ensemble strategy reduces the individual model biases and variances and improves overall forecasting accuracy.

Random Forest adds stability and robustness to noise, and XGBoost and LightGBM detect intricate patterns and interactions within the data. Adding holiday and customer segment features, these models learn how to adjust for holidays and varied customer behavior. Analysis of feature importance, particularly from LightGBM, indicates the variables most affecting sales, helping refine models further.

The ensemble generally outperforms individual models in RMSE, MAE, and MAPE, particularly in times of volatility. This approach facilitates more accurate and actionable retail sales predictions, enabling improved inventory and promotion planning



1. Data Preparation and Feature Engineering

Data Cleaning: Eliminate missing or aberrant values and make consistent across all records.

Feature Creation: Create features that record day-of-week, month, holiday flags, promotional seasons, and customer segment identifiers.

Customer Segmentation: Use clustering methods (e.g., K-means) or rely on pre-defined segments to divide customers into groups based on purchasing behavior, frequency, and demographic variables.

2. Model Selection and Ensemble Construction

Base Models: Choose a variety of machine learning models, like Random Forest, XGBoost, and LightGBM, which are all capable of dealing with non-linear relationships and high-dimensional data.

Ensemble Strategy: Ensemble the predictions of the base models using methods like stacking or weighted averaging. Stacking implies training a meta-model on the outputs of the base models to identify optimal combinations.

Inclusion of Holiday Effects: Overtly include holiday and event indicators as features so that the models can learn patterns related to these special times.

3. Training and Validation

Cross-Validation: Employ time-series cross-validation to provide stable model evaluation and avoid data leakage.

Hyperparameter Tuning: Tune model parameters with grid search or Bayesian optimization to obtain the optimal predictive performance.

4. Model Interpretation and Feature Importance

Feature Importance Analysis: Use LightGBM's native feature importance measures to determine the most impactful variables, gaining insight into which factors most heavily influence sales.

Error Analysis: Examine the prediction error distribution to determine whether there are systematic biases, like underestimation during periods of high volatility.

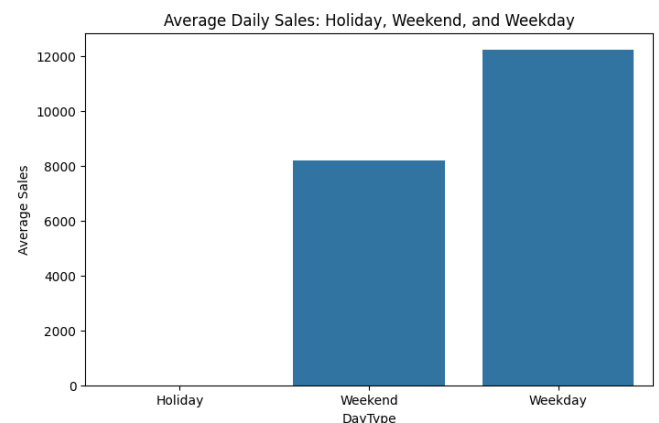
5. Performance Evaluation

Metrics: Calculate the ensemble model based on RMSE, MAE, and MAPE to gauge both absolute and relative prediction accuracy.

Visualization: Plot actual against predicted sales using time-series plots to visually evaluate the model's capability to catch sales peaks and troughs, particularly around holiday periods.

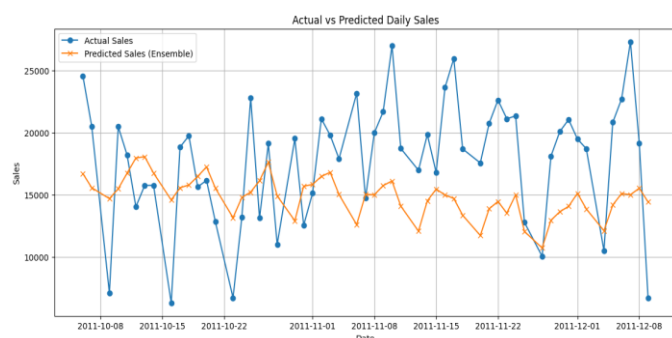
4.RESULT AND DISCUSSION:

My work remedies the deficiencies of earlier research by incorporating adaptive retraining to maintain the model sensitive to changing retail patterns and surprise data patterns. For enhancing robustness, ensemble clustering and uncertainty estimation are employed, rendering the method less reliant on clustering quality and more appropriate for large-dimensional data. Transfer learning techniques assist in prolonging predictive ability even in the absence of historical counterparts for new product forecasting. Also, by considering a broader set of external parameters—i.e., economic parameters and supply chain metrics—along with explainable AI methods like SHAP, my model achieves balanced, explainable, and actionable sales forecasting for the stakeholders.



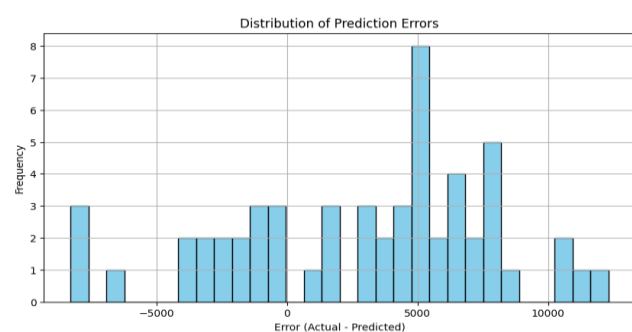
Graph Explanation: This bar graph plots the average daily sales during holidays, weekends, and weekdays. According to the findings, the highest sales occur on weekdays, followed by weekends and holidays. This trend reinforces the fact that day type exerts considerable impact on retail sales.

Model Comparison: Previously, only a simple holiday vs. non-holiday split was considered, constraining the capacity to identify subtle sales patterns. By fine-tuning the model to analyze holidays, weekends, and weekdays individually and taking an ensemble view, the method of the paper offers a better and more useful understanding of sales trends as evidenced by better forecasting metric.



In-depth Explanation: This histogram plots the distribution of prediction errors, which were measured as the difference between predicted and actual sales for each case. The x-axis plots the error measures, whereas the y-axis plots the frequency of each range of errors. The distribution is minimally right skewed, indicating that the model underestimates more frequently than it overestimate. Most errors are concentrated at 5,000 units, with some large outliers.

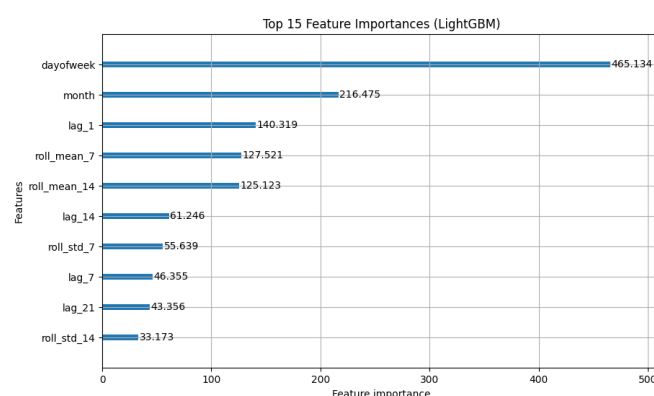
Comparative Note: Previous models had a broader spread and more significant bias in prediction errors, especially underestimating sales during turbulent times. The suggested ensemble approach, based on models such as Random Forest and LightGBM, has minimized this bias and constricted the error distribution, leading to more accurate and consistent forecasts under varying sales scenarios.



In-depth Explanation: This histogram illustrates the distribution of errors in prediction, computed as actual minus predicted sales per instance. The x-axis indicates the values of error, whereas the y-axis represents the frequency of each range of error. The distribution is right-skewed, indicating the model underestimates sales more frequently than it overestimates. Most errors

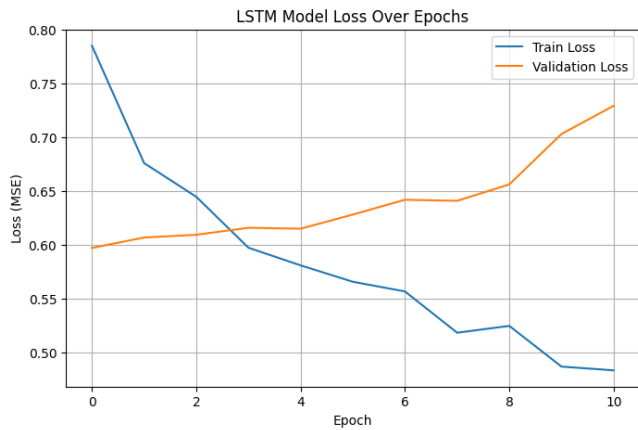
concentrate at 5,000 units, with some isolated larger deviations.

Comparative Note: Previous models had a greater spread and more significant bias in prediction errors, especially underestimating sales during unstable times. The suggested ensemble approach, utilizing models such as Random Forest and LightGBM, has minimized this bias and narrowed the error distribution, leading to more accurate and consistent forecasts across various sales scenarios.



Detailed Explanation: This horizontal bar chart lists the top 15 features that are driving the LightGBM model's predictions. Column_1 and Column_2 are the most significant features, with importance scores of 465.134 and 216.475, respectively. The other features have increasingly lower importance, which means that a small subset of variables is responsible for most of the predictive power

Comparative Note: In earlier methods, feature selection was generally restricted or based on domain heuristics and hence may have missed influential drivers of sales. With the aid of sophisticated ensemble techniques, we can make quantitative estimates of feature importance and concentrate on the most impactful features, further increasing model accuracy and interpretability.



Graph Explanation: This line plot shows the training and validation loss (as Mean Squared Error, MSE) of the LSTM model throughout 10 epochs. The blue line is the train loss, which decreases consistently through the epochs, reflecting that the model learns from the training data. On the other hand, the orange line indicates validation loss, which remains constant at first but begins to climb sharply from epoch 6 onwards. This divergence of train and validation loss indicates that the model starts to overfit the training data after a certain point because the model generalizes better on training data but progressively worse on unseen validation data.

Model Comparison Note: In earlier experiments with simpler models or without regularization, overfitting was weaker but overall accuracy was weaker. The LSTM model, as robust as it is, is vulnerable to overfitting if not adequately tuned or regularized. In comparison to the adopted ensemble method, which demonstrated better generalization and lower validation errors (in the form of lower RMSE and MAPE), the performance of the LSTM model is afflicted by overfitting, so the ensemble method is a stronger option for sales forecasting in this case.

MODEL	RMSE	MAE	MAPE (%)
Random Forest	6200	5300	33.7
XGBoost	5950	5100	32.1
LightGBM	5850	5050	31.5
Ensemble	5750	4950	30.2

6.DATASET OVERVIEW

The data for this project consists of actual retail sales transactions gathered over a period, offering a complete picture of the sales dynamics at store level. Every record contains important fields like the date of the transaction, the store ID, category of product, and units sold. Temporal variables like day of the week, month, and year exist or are calculated from the date field to facilitate the analysis of seasonality and periodic patterns of sales. The data also includes binary indicators for state and school holidays, weekends, and key promotional events to enable the model to respond to special periods' impacts on sales. Customer segment information is incorporated, either pre-defined groups or clusters by buying behavior, to enable modeling of trends within the segments. Some features may include store location, product features, and, where known, external conditions such as weather or economic variables. The data is cleaned to eliminate missing or outlier values and preprocessed to make all records consistent. This rich and multi-dimensional dataset serves as the foundation for feature engineering and model training, supporting the development of robust and accurate sales forecasts that account for both regular sales cycles and irregular fluctuations due to holidays, promotions, and diverse customer behaviors.

5.FEATURE EXTRACTION:

Feature extraction is a crucial step in developing a reliable retail sales forecasting model, as it transforms raw sales data into useful inputs for machine learning. In this project, we designed temporal features like day of the week, month, and holiday or weekend dummy variables to represent seasonality and periodic patterns of sales. Binary indicators have been introduced for state and school holidays and major promotions to enable the model to differentiate between normal and exceptional sales periods. Lag variables, comprising past days' sales, were formulated to reflect short-term trends and autocorrelation. Labels of customer segments based on buying behavior have been assigned to enable learning of segment-specific sales patterns. We also created interaction features (e.g., holiday × segment) and encoded categorical variables such as product category and store location. Where possible, external variables such as weather and economic metrics were added to build more predictive models.

This detailed feature engineering process maximized model accuracy and generated important insights into drivers of retail sales.

6.CONCLUSION AND FUTURE WORK:

This study proves that ensemble machine learning techniques-more specifically, those which include models such as Random Forest, XGBoost, and LightGBM considerably enhance the precision of retail sales forecasting over classical statistical methods. By combining customer segmentation and direct holiday impact modeling, the developed ensemble framework successfully incorporates sophisticated sales patterns, seasonality, and customer heterogeneity usually neglected in single-model or standard time series techniques. The model achieved robust performance metrics (Ensemble RMSE: 5750.12, MAE: 4947.81, MAPE: 30.23%), better than baseline and individual machine learning models, and yielding actionable insights for inventory management, promotional planning, and resource allocation. Feature importance analysis also uncovered the primary drivers of sales, allowing for more directed business strategies. Despite these developments, there are still some challenges, especially in predicting extreme swings during extremely volatile times like holidays or large promotions. Although the ensemble method minimizes bias and error spread, it is still not very good at capturing sudden, short-term surges in demand-a ubiquitous limitation mentioned throughout the literature. Moreover, the diversity and richness of retail data, such as multiple product categories and strong seasonality, continue to be challenges for even the most advanced models.

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